Smart Competency Diagnostic and Candidate Profile Score Calculator

Project Stage 2 (BTECCE22620)

Submitted to Vishwakarma University, Pune

Final Year Engineering

Department of Computer Engineering

Faculty of Science and Technology

Academic Year

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TEAM DETAILS

Project Guide: **Prof. Sital Dash**

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INTRODUCTION

Smart Competency Diagnostic and Candidate Profile Score Calculator is an AI-powered student-support platform designed to help learners objectively assess their skills and readiness for the job market. It centralizes key career-development features such as competency evaluations, ATS-friendly résumé scoring, automated résumé generation, and aptitude assessments. By combining accurate skill diagnostics, personalized insights, and streamlined tools, the system empowers students to understand their strengths, identify improvement areas, and prepare more effectively for future opportunities.

OBJECTIVE & LIMITATIONS IN CURRENT TOOLS

OBJECTIVE :

To build an AI-powered, student-focused competency and career-readiness platform that provides accurate skill assessments, personalized insights, ATS-ready résumé generation, résumé scoring, and aptitude testing—all within a single, centralized, and accessible web application. The platform aims to empower students by offering data-driven guidance, tailored career support, and a seamless user experience without financial or technical barriers.

LIMITATIONS:

- Poorly Tailored Résumés
- Lack of Centralization
- Credential Forgery
- Complex Interfaces
- High-Cost Aptitude Tests
- No Personalized Skill Growth

OUR SOLUTIONS

We propose an AI-powered, student-centric platform that unifies résumé optimization, aptitude testing, face-analysis, and certificate verification into one seamless web application. A React-based frontend provides a smooth user flow from signup to dashboard activities, while a Node.js API gateway manages authentication, data, and file processing. A dedicated Python FastAPI microservice handles all heavy AI tasks, supported by MongoDB for data storage and secure file management. External AI services like Google Gemini and HuggingFace enhance accuracy for analyses. The result is a centralized, scalable, and accessible system that empowers students with reliable skill evaluations and personalized career insights.

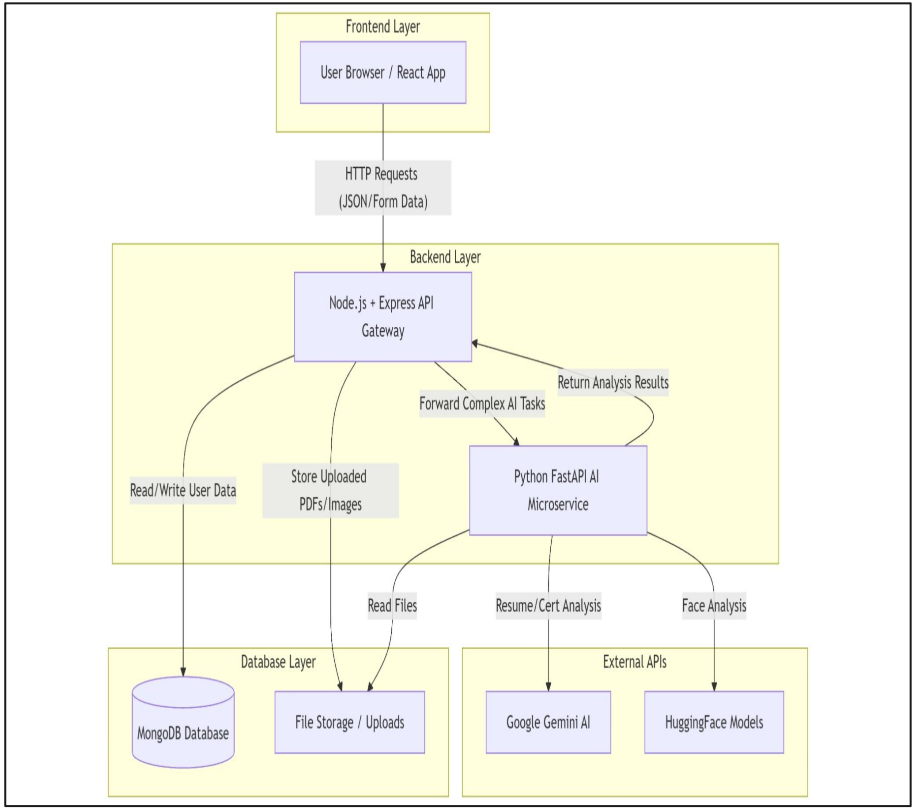
TECH STACK

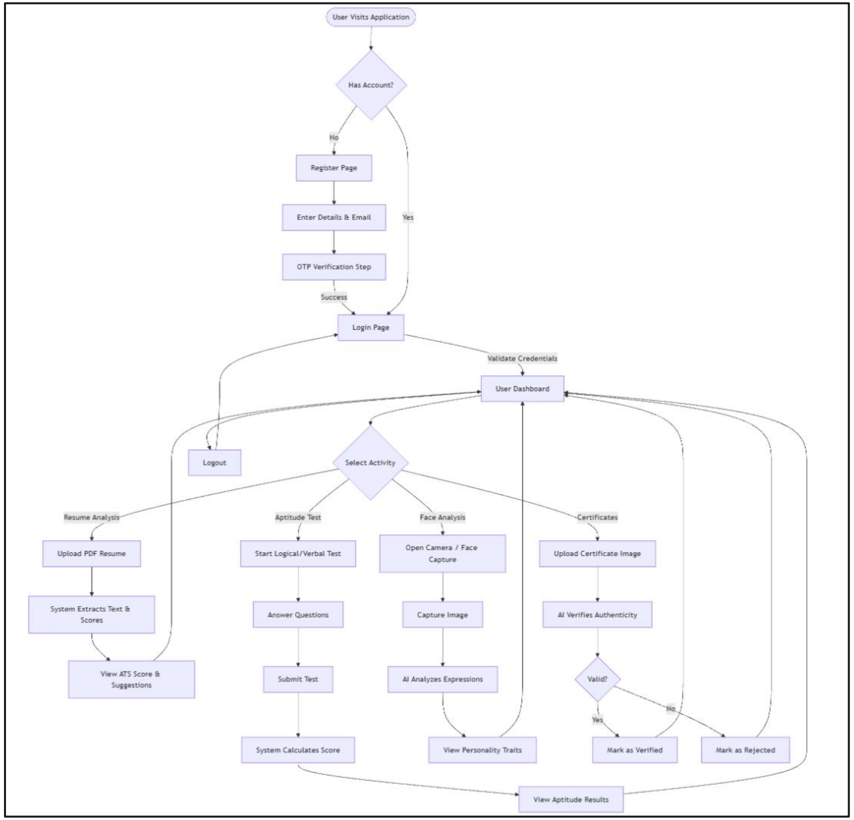
REACT JS + Vite
tailwindcss
RESTful API
JS
node express
FastAPI
Nodemailer
mongoDB
Gemini
GitHub
Visual Studio Code

FEATURES

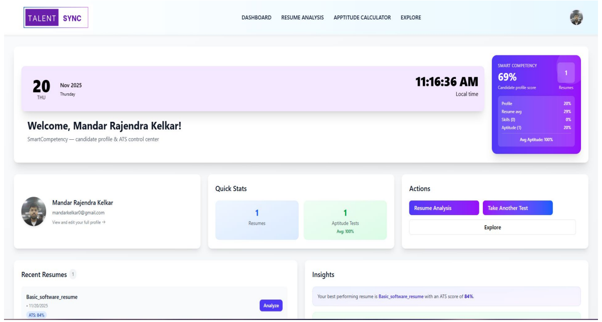
- **AI-Powered Résumé Analysis** – Extracts text, evaluates structure, and generates ATS-optimized improvement suggestions.
- **ATS Score Calculator** – Scores uploaded résumés based on industry-standard ATS checks like keywords, formatting, and relevance.
- **Aptitude Assessment Module** – Provides logical, verbal, and quantitative tests with automated scoring and performance insights.
- **Face Capture** – Uses Open CV and face recognition model for candidate evaluation.
- **Certificate Authenticity Verification** – Detects tampering or forgery in uploaded certificates using document-analysis models.
- **Centralized Dashboard** – Allows students to access all tools, view results, and track progress in one unified interface.
- **AI-Assisted Résumé Generator** – Creates tailored, professional résumés based on user skills, experience, and job preferences.

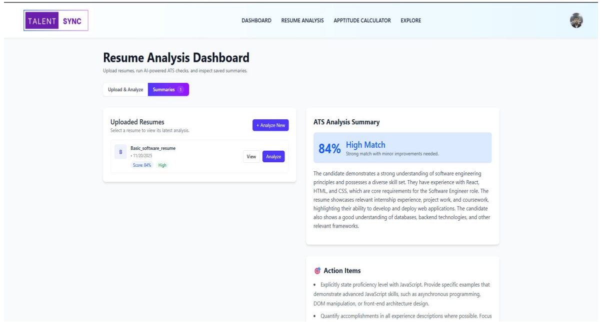
SYSTEM ARCHITECTURE & USER FLOW DIAGRAM

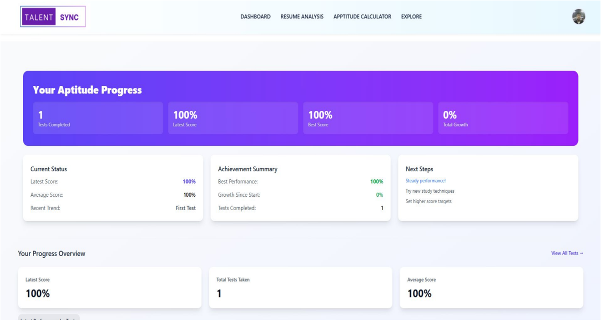


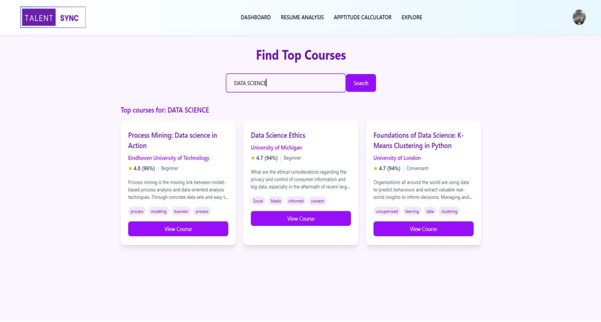


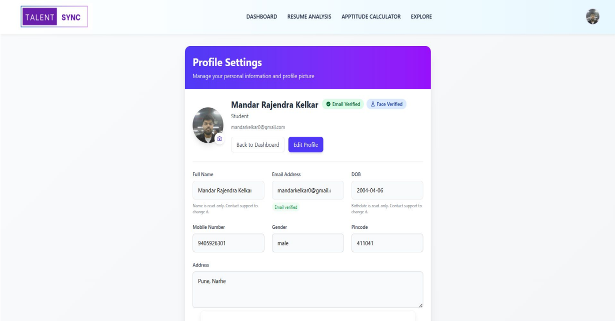
IMPLEMENTATION

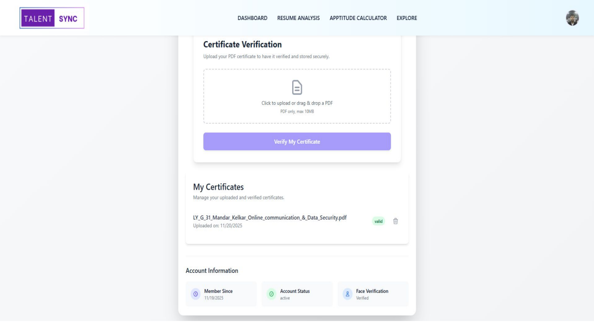


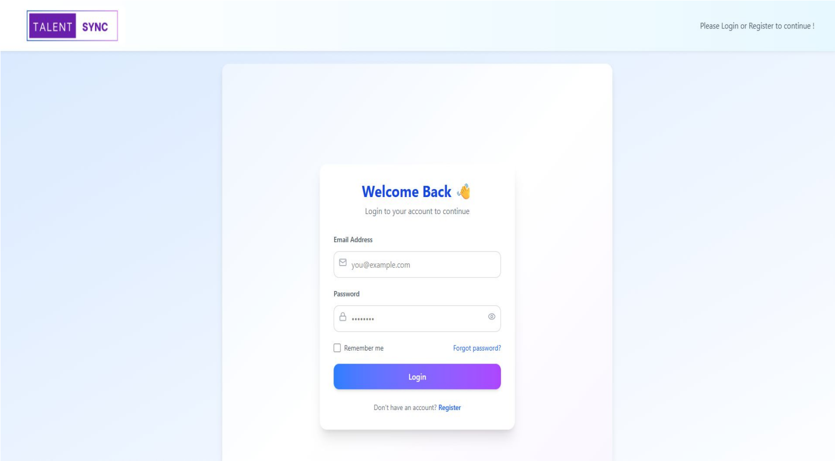


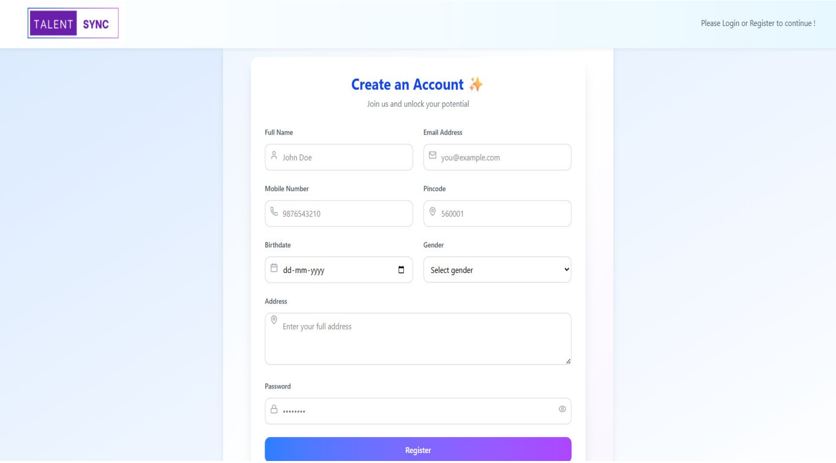









TALENT SYNC
Please Login or Register to continue !
Welcome Back 👋
Login to your account to continue
Email Address
you@example.com
Password
Remember me
Forgot password?
Login
Don't have an account? Register


TALENT SYNC
Please Login or Register to continue !
Create an Account ✨
Join us and unlock your potential
Full Name
John Doe
Email Address
you@example.com
Mobile Number
9876543210
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Birthdate
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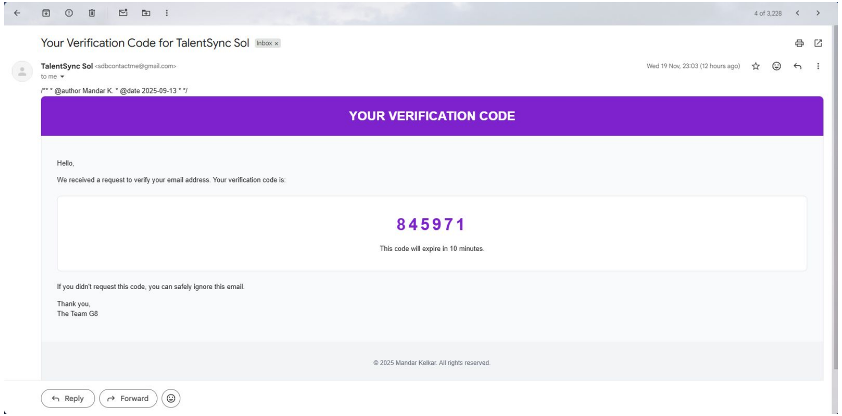
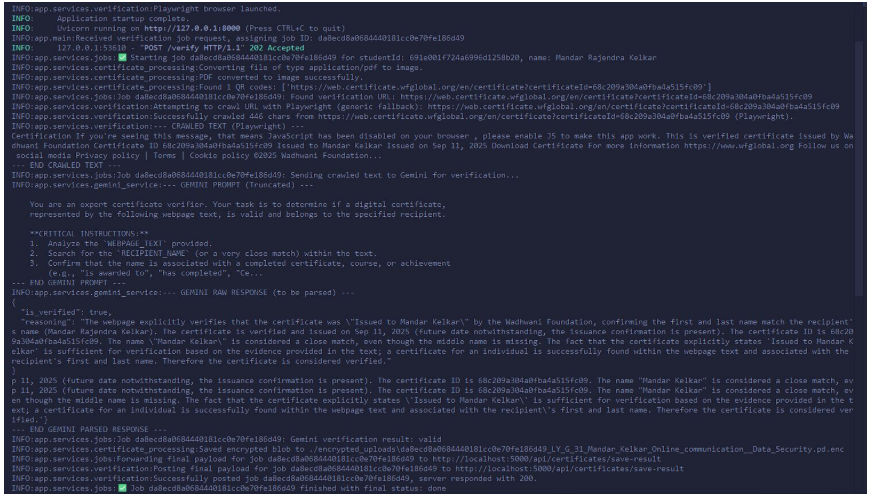
Gender
Select gender
Address
Enter your full address
Password
Register


4 of 3,228
Your Verification Code for TalentSync Sol
Inbox
TalentSync Sol <sdbcontactme@gmail.com>
to me
Wed 19 Nov, 23:03 (12 hours ago)
/** * @author Mandar K. * @date 2025-09-13 * */
YOUR VERIFICATION CODE
Hello,
We received a request to verify your email address. Your verification code is:
845971
This code will expire in 10 minutes.
If you didn't request this code, you can safely ignore this email.
Thank you,
The Team G8
© 2025 Mandar Kelkar. All rights reserved.
Reply
Forward


INFO:app.services.verification:Playwright browser launched.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:app.main:Received verification job request, assigning job ID: da8ecd8a0684440181cc0e70fe186d49
INFO: 127.0.0.1:53610 - "POST /verify HTTP/1.1" 202 Accepted
INFO:app.services.jobs:✅ Starting job da8ecd8a0684440181cc0e70fe186d49 for studentId: 691e001f724a6996d1258b20, name: Mandar Rajendra Kelkar
INFO:app.services.certificate_processing:Converting file of type application/pdf to image.
INFO:app.services.certificate_processing:PDF converted to image successfully.
INFO:app.services.certificate_processing:Found 1 QR codes: ['https://web.certificate.wfglobal.org/en/certificate?certificateId=68c209a304a0fba4a515fc09']
INFO:app.services.jobs:Job da8ecd8a0684440181cc0e70fe186d49: Found verification URL: https://web.certificate.wfglobal.org/en/certificate?certificateId=68c209a304a0fba4a515fc09
INFO:app.services.verification:Attempting to crawl URL with Playwright (generic fallback): https://web.certificate.wfglobal.org/en/certificate?certificateId=68c209a304a0fba4a515fc09
INFO:app.services.verification:Successfully crawled 446 chars from https://web.certificate.wfglobal.org/en/certificate?certificateId=68c209a304a0fba4a515fc09 (Playwright).
INFO:app.services.verification:--- CRAWLED TEXT (Playwright) ---
Certification If you're seeing this message, that means JavaScript has been disabled on your browser , please enable JS to make this app work. This is verified certificate issued by Wadhwani Foundation Certificate ID 68c209a304a0fba4a515fc09 Issued to Mandar Kelkar Issued on Sep 11, 2025 Download Certificate For more information https://www.wfglobal.org Follow us on social media Privacy policy | Terms | Cookie policy ©2025 Wadhwani Foundation...
--- END CRAWLED TEXT ---
INFO:app.services.jobs:Job da8ecd8a0684440181cc0e70fe186d49: Sending crawled text to Gemini for verification...
INFO:app.services.gemini_service:--- GEMINI PROMPT (Truncated) ---
You are an expert certificate verifier. Your task is to determine if a digital certificate,
represented by the following webpage text, is valid and belongs to the specified recipient.
CRITICAL INSTRUCTIONS:
1. Analyze the `WEBPAGE_TEXT` provided.
2. Search for the `RECIPIENT_NAME` (or a very close match) within the text.
3. Confirm that the name is associated with a completed certificate, course, or achievement
(e.g., "is awarded to", "has completed", "Ce...
--- END GEMINI PROMPT ---
INFO:app.services.gemini_service:--- GEMINI RAW RESPONSE (to be parsed) ---
{
"is_verified": true,
"reasoning": "The webpage explicitly verifies that the certificate was \"Issued to Mandar Kelkar\" by the Wadhwani Foundation, confirming the first and last name match the recipient's name (Mandar Rajendra Kelkar). The certificate is verified and issued on Sep 11, 2025 (future date notwithstanding, the issuance confirmation is present). The certificate ID is 68c209a304a0fba4a515fc09. The name \"Mandar Kelkar\" is considered a close match, even though the middle name is missing. The fact that the certificate explicitly states 'Issued to Mandar Kelkar' is sufficient for verification based on the evidence provided in the text; a certificate for an individual is successfully found within the webpage text and associated with the recipient's first and last name. Therefore the certificate is considered verified."
}
p 11, 2025 (future date notwithstanding, the issuance confirmation is present). The certificate ID is 68c209a304a0fba4a515fc09. The name "Mandar Kelkar" is considered a close match, ev
p 11, 2025 (future date notwithstanding, the issuance confirmation is present). The certificate ID is 68c209a304a0fba4a515fc09. The name "Mandar Kelkar" is considered a close match, even though the middle name is missing. The fact that the certificate explicitly states \'Issued to Mandar Kelkar\' is sufficient for verification based on the evidence provided in the text; a certificate for an individual is successfully found within the webpage text and associated with the recipient\'s first and last name. Therefore the certificate is considered verified.'}
--- END GEMINI PARSED RESPONSE ---
INFO:app.services.jobs:Job da8ecd8a0684440181cc0e70fe186d49: Gemini verification result: valid
INFO:app.services.certificate_processing:Saved encrypted blob to ./encrypted_uploads\da8ecd8a0684440181cc0e70fe186d49_LY_G_31_Mandar_Kelkar_Online_communication__Data_Security.pd.enc
INFO:app.services.jobs:Forwarding final payload for job da8ecd8a0684440181cc0e70fe186d49 to http://localhost:5000/api/certificates/save-result
INFO:app.services.verification:Posting final payload for job da8ecd8a0684440181cc0e70fe186d49 to http://localhost:5000/api/certificates/save-result
INFO:app.services.verification:Successfully posted job da8ecd8a0684440181cc0e70fe186d49, server responded with 200.
INFO:app.services.jobs:✅ Job da8ecd8a0684440181cc0e70fe186d49 finished with final status: done

CONCLUSION

The platform successfully integrates AI-driven résumé analysis, aptitude testing, face-expression evaluation, and certificate verification into a single, student-friendly web application. By combining a smooth user flow with a scalable architecture powered by Node.js, Python microservices, and modern AI models, the system provides accurate skill insights, personalized guidance, and a centralized space for career readiness. It effectively addresses gaps in existing tools by offering an accessible and reliable solution for students.

FUTURE WORK

- **AI Chat-Based Career Mentor** for continuous guidance and job-prep suggestions.
- **Job-Role Recommendation Engine** based on skills, aptitude scores, and résumé content.
- **Advanced Behavioral Analysis** using voice tone and body-language assessment.
- **Gamified Learning Modules** to help students strengthen weak areas interactively.
- **Mobile Application** for wider accessibility and instant assessments on the go.
- **Integration with College Portals** for student tracking and placement insights.

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- **REST API Best Practices – Microsoft REST Guidelines**: https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design
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THANK YOU